

Background Scene Dominant Color based Visibility Enhancement of Head-Up Display

Hyuk Jun Yoon, Yongwan Park

Dept. of Information and Communication Engineering
Yeungnam University
Gyeongsan, Gyeongbuk, Republic of Korea
hkjun4760@ynu.ac.kr, ywpark@yu.ac.kr

Ho-Youl Jung*

Dept. of Information and Communication Engineering
Yeungnam University
Gyeongsan, Gyeongbuk, Republic of Korea
hoyoul@yu.ac.kr

*Corresponding Author

Abstract— Images projected from HUD (Head-up Display) are always overlaid with moving background scene. In such cases, drivers suffer from understanding instinctively the HUD information. Visibility enhancement of HUD information is very important for drivers to recognize easily and quickly the information. In this paper, we propose two visibility enhancement methods of HUD image, which are based on the dominant color of background scene. One is to change the color of symbols, another is to add different color outline to symbols of HUD information, depending on the similarity between HUD information and background dominant colors. To compare the color similarity, HSI (Hue, Saturation, Intensity) color domain is used. The proposed methods are designed to minimize the color change, so as to maintain the meaning of symbol color. Through the simulations, we show that the proposed color changing and outline methods are effective to improve the instinctive understanding.

Keywords—HUD; Head-Up Display; Visibility; Image Processing;

I. INTRODUCTION

When driving, driver should look forward. This is the reason why HUD (Head-Up Display HUD) has been widely installed in many vehicle models. HUD is display system that presents some important driving information through windshield, without requiring drivers to look away from their usual viewpoints. Drivers are able to view the information with looking forward, instead of angled down looking at lower dashboard. HUD also has the advantage that the driver's eye do not need to refocus to view the outside after looking at the optically nearer instruments [1]. In recent, advanced light sources such as LED and laser diode allow to project bigger size image on the plane longer distance from driver's eye. In particular, the light sources permit enhancing the visibility in both true color and brightness. However, images projected from HUD are always overlaid with real scene, i.e., moving background scene. If the symbols displayed from HUD have the same color as background scene, drivers can suffer from understanding instinctively the HUD information. It is clear that the visibility has to be enhanced by considering the background scene. The visibility of the HUD image should be

guaranteed under various environments, so that drivers can recognize the HUD information immediately.

Sometimes, the color of the HUD images looks like different, according to the background. Visibility can be degraded in the following cases; suddenly around the darkened or brightened, light reflected due to streetlight, and etc.

In this paper, we propose visibility enhancement methods using dominant color of background scene. One is to change the color of HUD information, and another is to add different colored outline to HUD information according to the background color.

The paper outline is as follows. In Section II, we introduce the existing methods to improve the visibility of HUD. In Section III, two proposed methods are described. In Section IV, we show the simulation results using the proposed algorithm. Finally, the conclusion is shown in In Section V.

II. RELATED WORKS

A. HUD Color

It is important to determine the color of symbols, which are projected by HUD. The colors should be selected to quickly deliver drivers the information. In [2], the authors analyze driver's EGG signal when driver looking at certain color symbol. They insist that green, yellow and orange are suitable to represent HUD information. In commercial HUD systems, red, blue, yellow and white colors have been often used. For example, it is known that yellow color is used in more than 70% of the commercial HUD products [3].

B. HSI Color Domain

Even though the color of symbol is well determined, it is necessary to consider colors of the real background scene. The HUD image is displayed with overlapping the background. If background has the same color or very similar to the color of symbols, drivers cannot instinctively recognize the HUD information.

Some research results are reported to enhance the visibility of the HUD image depending on the background scene. The background color is measured by using a camera and compared with the color of the image to be projected by HUD.

If the color to be projected is the same or similar within a certain range, it is changed to other.

In general, color image is represented by RGB format. For efficient representation of the color, the RGB color often transforms to HSI (Hue, Saturation, Intensity) format. Here, H component represents the color. Where, RGB values have the dynamic range [0, 1]. The input RGB format image is transformed to HSI format as given by

$$\begin{aligned}\theta &= \cos^{-1} \left[\frac{\frac{1}{2}[(R-G) + (R-B)]}{\sqrt{(R-G)^2 + (R-B)(G-B)}} \right] \\ H &= \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{otherwise} \end{cases} \\ S &= 1 - \frac{\min(R, G, B)}{(R+G+B)} \\ I &= \frac{1}{3}(R+G+B)\end{aligned}\quad (1)$$

Where H component has the dynamic range [0, 360], Saturation and Intensity have the dynamic range [0, 1]. The HSI format is inversely transformed to RGB format as follows [3].

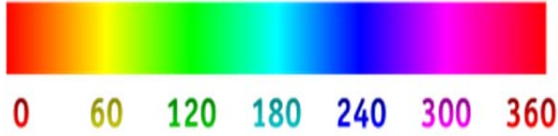


Figure 1. Color of hue (H) component

$$\begin{aligned}\text{RG sector } (0^\circ \leq H < 120^\circ) \\ R &= I \left[1 + \frac{\cos H}{\cos(60^\circ - H)} \right], G = 1 - (R + B), \\ B &= I(1 - S) \\ \text{GB sector } (120^\circ \leq H < 240^\circ) \\ H &= H - 120^\circ \\ R &= I(1 - S), G = I \left[1 + \frac{\cos H}{\cos(60^\circ - H)} \right], \\ B &= 1 - (R + G) \\ \text{BR sector } (240^\circ \leq H < 360^\circ) \\ H &= H - 240^\circ \\ R &= 1 - (G + B), G = I(1 - S), \\ B &= I \left[1 + \frac{\cos H}{\cos(60^\circ - H)} \right]\end{aligned}\quad (2)$$

C. Complementary Color Algorithm

Even though the color of symbol is well determined, it is necessary to consider colors of the real background scene. The HUD image is displayed with overlapping the background. If background has the same color or very similar to the color of symbols, drivers cannot instinctively recognize the HUD information.

Some research results are reported to enhance the visibility of the HUD image depending on the background scene. The

background color is measured by using a camera and compared with the color of the image to be projected by HUD. If the color to be projected is similar to background, it is changed to other color.

If background and HUD projection image have similar color, the color of HUD image changes to the complementary color of background [5]. The complementary color is obtained by rotating 180° from the original H component in HIS domain and transforming it to RGB format. The HUD image displays the information using the converted RGB value. This method is helpful to increase the visibility of HUD image according to varying background. However, abrupt color change can cause the confusion of drivers. In particular, the complementary color changing method eliminates the inherent meaning of color for each symbol in HUD image. For example, it is well known that red and yellow indicate danger and attention respectively.

D. Color Correction for Mobile Beam Projector

In the field of mobile beam projector, it is necessary to develop the color correction technology considering the color of projection plane. In [6], beam projector color correction technique is introduced. The method applies in the RGB color space and consists of the following steps. First, spatial mapping performs using down-sampling. In the second step, luminance of the original image to be projected is compensated. In third step, chromaticity is corrected by using color invariant model. For luminance compensation, the original RGB image was transformed into the YCbCr color format to extract the luminance values. Then, the luminance value is corrected to preserve the luminance distribution of the original image on the colored screen. In the chromaticity correction, the chromaticity triplet of the image was converted by inverse YCbCr transformation. The color is adjusted based on the chromaticity invariant representation to reduce the influence of modulation in the appearance of the projected image. This method is efficient to preserve the original color to be projected even when colored (background) screen is used. However, the method cannot apply to HUD application where the projected image should be distinct from the background.

III. PROPOSED ALGORITHM

As mentioned before, the complementary color change is helpful to distinguish the HUD information from the background. Then, drivers can recognize the HUD information easily. However, the color is also very important information for drivers to intuitively understand the situation of driving environment such as danger and attention. Drivers cannot catch cautious situation, if danger sign displays with complementary color of red, i.e. blue as shown in Figure 1. Drivers cannot recognize the exact meaning of the HUD information. It is clear that the original red color should be kept as possible.

In this paper, we propose two visibility enhancement methods for HUD projection image. One is to change the color of symbols, another is to add different colored outline to symbols of the HUD information. The proposed methods

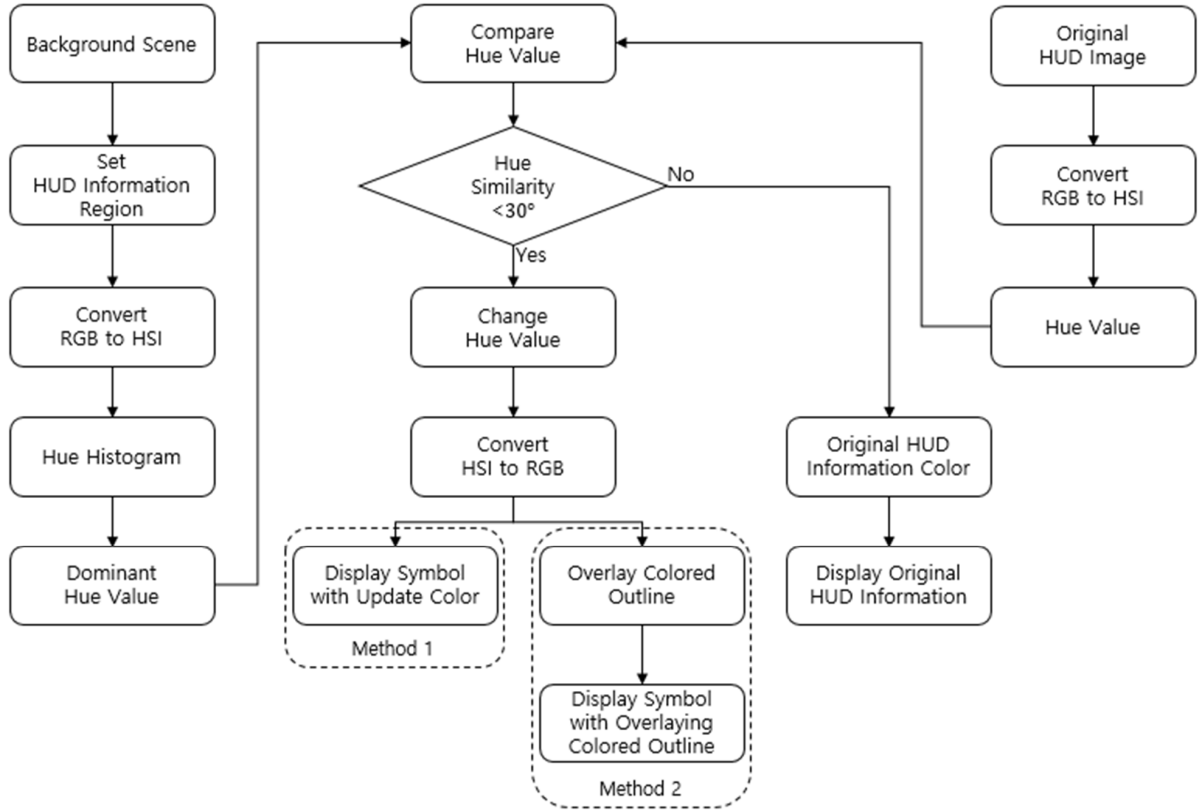


Figure 2. Proposed Algorithm

estimate the similarity of color between HUD image and real driving environment (background). Background image is captured by a forward-facing camera mounted behind the windshield of the host vehicle. Dominant color is obtained in each region of background image by using histogram of H component in HIS domain. The dominant color is compared with the original color of symbol to be projected.

Figure 2 shows the procedure of the proposed two methods, where method 1 and method 2 have only different in display the updated color. First, we assume that the background scene captured by camera covers enough region of image originally projected by HUD, and the two images be matching on the same coordinate. As background capturing camera is positioned out of eye-box of HUD, it takes only background scene and does not prevent drivers from driving. The region of HUD image is determined in the camera input image. The two RGB format images are converted to HIS ones, and H component is analyzed in the next steps. Dominant color is estimated in each local region of background image by using H component histogram. In this paper, the maximum frequency color is determined as dominant color. H component of symbol to be projected in the same region is also obtained. The two H components are compared. If they are closer than 30 degree, original color of symbol is up-dated to other one. In the method 1, the symbol color is changed to the up-dated color. In method 2, the up-dated colored outline is overlaid to the symbol, the original color maintains in the

symbol. If the difference between two H components is bigger than 30 degree, the original color of the HUD image maintains as it is distinct from the background. There is no change of color and no overlaying outline in both methods.

A. Use HSI Color Domain

In the real driving situation, the forward background image is captured by using the camera. The color is compared and changed using the Hue value of the HSI color domain. Both background input image and HUD image represent in RGB color format. The images are transformed into HIS format by using equation (1). Only H component is up-dated according to the similarity, while maintaining original S and I components. The up-dated H component with original S and I components are inversely converted to RGB format by using equation (2). The changed RGB format color is applied in both proposed method.

B. Hue Component Histogram & Dominant Hue Value

In the input background image, the ROI (Region of Interest) is set as the position where the HUD information displays. After RGB to HIS conversion, H component Histogram is obtained and used to find maximum frequency H component. The H component is the dominant color of background in certain local region.



Figure 3. Simulation results displaying by (a) original HUD color information color, (b) up-dated color using method 1, and (c) overlaying outline using method 2

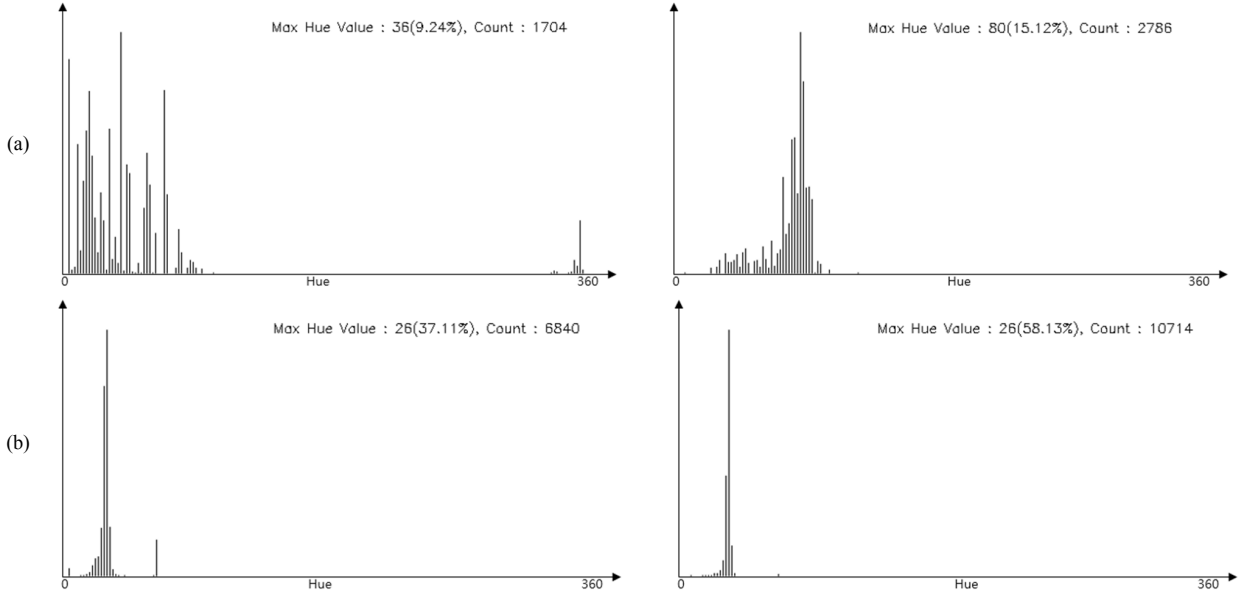


Figure 4. Hue component histogram of (a) general background with red color region (left) and with yellow color region (right graph), (b) background image shown in Figure 3 with left region (left) and right region (right)

C. Change Hue Component of HUD image

The dominant H component of background image is compared with the H component of symbol to be displayed by HUD. If the difference D between two H components is measured by

$$D = |H_{hud} - H_D| \quad (3)$$

Where H_{hud} indicates the original H component of HUD image and H_D indicates the dominant H component of background image. If the difference D is smaller than 30° , it is determined that HUD information and the background image have similar color to each other. In this case, Hue component of the HUD information is changed to have a bigger difference as follows;

$$H_{new_hud} = \begin{cases} H_{hud} + (30^\circ - D), & \text{if } H_{hud} > H_D \\ H_{hud} - (30^\circ - D), & \text{otherwise} \end{cases} \quad (4)$$

If the difference is bigger than 30° , the Hue component of the HUD information is not changed.

D. Method 1

HUD symbol color is changed by the new H component H_{new_hud} . The H_{new_hud} with original S and I values are converted to RGB color format and displayed by HUD.

E. Method 2

HUD symbol is highlighted by outline of which color is the H_{new_hud} . This method allows maintaining the original color of the symbol by overlaying the colored outline. It means that the original HUD information meaning can keep

on. If the difference D is bigger than 30° , the original symbol displays without outline.

IV. SIMULATION RESULT

The proposed methods are tested on various video sequences which are captured by camera facing forward. This simulation performs on PC to estimate the performance of the proposed method. The HUD symbol represented by up-dated color is displayed on the monitor overlaying the background video. Figure 3 shows the simulation results displaying by original HUD color information color, up-dated color using method 1, and overlaying outline using method 2. Figure 4 shows the histogram of H component in background scene, where the maximum frequency indicates the dominant H component of background.

Figure 3 shows that the proposed methods are very helpful to enhance the visibility of HUD image according to the background color. In particular, the method 2 has better performance of visibility, maintain the original color within the symbol.

V. CONCLUSION

The HUD test will be tested with the HUD simulator, but we confirmed the results on PC because of HUD equipment and simulator implementation problem. But we confirmed that the visibility was improved by applying our proposed methods that comparing the Hue value of the HUD information after measuring the dominant Hue value.

In this paper, we propose two visibility enhancement methods of HUD image, which are based on the dominant color of background scene. The proposed methods are designed to minimize the color change, so as to maintain the

meaning of symbol color. Through the simulations, we show that the proposed outline method is more effective to improve the visibility. For the future, the proposed methods will be tested on real driving environments or on driving simulators. We have a plan to perform simulation tests which satisfy ISO 9241-3(Visual Display Requirements) [7].

ACKNOWLEDGMENT

This work was supported by the Technology Innovation Program (No. 10063444, Development of Vehicle-Mounted HUD(Head Up Display) System with Laser Projection) funded By the Ministry of Trade, industry & Energy(MI, Korea).

This research was supported by the MSIP(Ministry of Science, ICT and Future Planning), Korea, under the ITRC(Information Technology Research Center) support program (IITP-2017-2016-0-00313) supervised by the IITP(Institute for Information & Communications Technology Promotion).

REFERENCES

- [1] Wikipedia; HUD (Head-Up Display)
- [2] Byeong-Sup Moon, Feng Li, Bum-Jin Park, and Jin-Nyung Heo, "HUD Information analysis using Driving Simulator," ICCV, vol. 2011, pp. 183–184, December 2011.
- [3] Min Ji Kim, and Young Hwan Pan, "A Study on the relationship between cognitive level and size according to the importance of vehicle Head-Up Display channeled information," Journal of Korea Design Knowledge, vol. 33, pp. 163–170, March 2015.
- [4] Gonzalez Woods, "Digital Image Processing", 2nd edition
- [5] Ming-Shou An, and Dae-Seong Kang, "An Implementation of Interpolation Algorithm and Complementary Color Algorithm for HUD," Journal of Korean Institute of Information Technology, vol.7, pp. 83–88, April 2009.
- [6] Chang-Hwan Son, and Yeong-Ho Ha, "Color Correction of Images Projected on a Colored Screen for Mobile Beam Projector," Journal of Imaging Science and Technology, vol.52, Num.3, pp. 30505-1–30505-11, May 2008.
- [7] ISO 9241-3:1992, Ergonomic requirements for office work with visual display terminals—Part 3:Visual display requirements